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Method and system for vertical gradient freeze 8 inch gallium arsenide substrates

Abstract

Methods and wafers for vertical gradient freeze 8 inch gallium arsenide (GaAs) substrates. In disclosed examples, vertical gradient freeze systems for forming gallium arsenide (GaAs) substrates having silicon as a dopant, the system includes a crucible to contain a GaAs liquid melt and seed material during a formation process; one or more heating coils arranged in a plurality of heating zones; and a pedestal to move relative to the crucible, the system operable to control heating of the plurality of heating zones and movement of the pedestal to form a single crystal GaAs substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS/INCORPORATION BY REFERENCE

(1) This application is a Divisional of U.S. patent application Ser. No. 17/700,746 filed on Mar. 22, 2022, which claims priority to and the benefit of U.S. Provisional Application No. 63/164,378 filed on Mar. 22, 2021, which is hereby incorporated herein by reference in its entirety.

FIELD

(1) Certain embodiments of the disclosure relate to semiconductor substrates. More specifically, certain embodiments of the disclosure relate to a method and system for vertical gradient freeze 8 inch gallium arsenide (GaAs) substrates.

BACKGROUND

(2) Semiconductor substrates, and specifically Group III-V semiconductor substrates, are used in the manufacture of electronic and optoelectronic devices such as light-emitting diodes (LEDs), lasers, heterojunction bipolar transistors (HBTs), and pseudo-morphic high-electron mobility transistors (pHEMTs), for example. Defects in the substrates can reduce yields and increase costs. (3) Further limitations and disadvantages of conventional and traditional approaches will become apparent to one of skill in the art, through comparison of such systems with the present disclosure as set forth in the remainder of the present application with reference to the drawings.

BRIEF SUMMARY

(4) A system and/or method for vertical gradient freeze 8 inch gallium arsenide (GaAs) substrates, substantially as shown in and/or described in connection with at least one of the figures, as set forth more completely in the claims. (5) Various advantages, aspects and novel features of the present disclosure, as well as details of an illustrated embodiment thereof, will be more fully understood from the following description and drawings.

Description

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating a vertical gradient freeze reactor, in accordance with an example embodiment of the disclosure. (2) FIG. 2 illustrates an 8 inch GaAs ingot and head and tail end faces of the ingot, in accordance with an example embodiment of the disclosure. (3) FIGS. 3A-3D illustrate etch pit density results for 8 inch gallium arsenide wafers manufactured in accordance with an embodiment of the disclosure. (4) FIGS. 4A-4F illustrate photoluminescence measurements of 8 inch gallium arsenide substrates, in accordance with an example embodiment of the disclosure. (5) FIGS. 5A-5F illustrate resistivity maps of 8 inch gallium arsenide substrates, in accordance with an example embodiment of the disclosure. (6) FIG. 6 illustrates a method for fabricating GaAs wafers using the vertical gradient freeze furnace 100 shown in FIG. 1. (7) FIG. 7 illustrates devices fabricated on low etch pit density 8 inch gallium arsenide wafers, in accordance with an example embodiment of the disclosure.

DETAILED DESCRIPTION

(8) Certain aspects of the disclosure may be found in a method and system for vertical gradient freeze 8 inch gallium arsenide (GaAs) substrates.

(9) FIG. 1 is a diagram illustrating a vertical gradient freeze reactor, in accordance with an example embodiment of the disclosure. Referring to FIG. 1, there is shown vertical gradient freeze (VGF) system **100** comprising an ampoule **110**, heating coils **120**, crucible **130**, and pedestal **140**. Within the ampoule **110**, growth may proceed using a seed crystal **101** and a liquid melt **105** in a vertical configuration, where the solid **103** forms due to very gradual cooling of the melt **105** with a boron trioxide B.sub.2O.sub.3 layer as an encapsulant on the melt **105**. The crucible **130** may comprise pyrolytic boron nitride (PBN), for example, and may contain the melt and seed materials during the process.

(10) The VGF system **100** comprises a configuration for growth charge, heater, insulation and pedestal **140** that includes: direct radiation from heating coils **120** above the level of cone in the crystal growth charge, the seed **101**, and may also comprise a pedestal **140** for support of the ampoule **110** in the heating coils **120**. In addition, insulation may be placed within the pedestal **140** to promote radial heat flux inward during growth and heat removal during post-growth process and cooling. The pedestal **140** for the growth ampoule **110** may comprise a hollow core below the seed pocket holding the seed crystal **101** to provide a conducting heat flow path downward through the center and further aided and made more stable by a larger diameter quartz rod down to the bottom of the heater. The pedestal **140** may be operable to move vertically to assist in the growth process, or remain at a fixed height. In addition, the pedestal **140** may rotate to provide further thermal uniformity during heating.

(11) In the VGF process, a polycrystal charge, the seed **101**, in crucible **130** in the ampoule **110** may be heated with a multi-zone furnace, as represented by the heating coils **120**. In an example scenario, a B.sub.2O.sub.3 encapsulant **107** may be placed on top of the melt to reduce loss of crystal growth material, particularly arsenic with higher vapor pressure. Crystal growth is initiated by melting the charge to the seed **101**, thereby forming the melt **105**, and slowly reducing the temperature to cool the melt **105** at the seed **101**. The crystallization process, forming the solid **103**, may be precisely controlled by varying the temperature gradient in different zones of the furnace, such as by applying different currents to different portions of the heating coils **120** and/or moving the pedestal **140** vertically. In addition, the pedestal **140** may rotate during heating and cooling to average out any thermal variations within the heating coils **120**. Accordingly, the heating coils **120** may comprise a multi-zone heating system where different temperatures, heating/cooling rates, and spatial temperature profiles are enabled.

(12) The VGF process has several advantages over other processes such as liquid encapsulated Czochralski (LEC) and horizontal Bridgman (HB). First, the thermal environment of the crystal and melt are radially symmetric, enabling a radially uniform interface and precise control of temperature profiles. Second, growth in low axial and radial temperature gradients, combined with diameter control imposed by a crucible, and no need to use a large temperature gradient to freeze the crystal quickly, significantly reduces the strain and, hence, dislocations in the crystals. Third, the liquid-solid interface progresses upward through the melt, with crystallization starting at the seed **101** in the bottom of the crucible **130**. The system **100** may be cooler at the bottom, and thermally stabilized against convection. Even in the cooling process, easily adjustable cooling brings the crystal to room temperature at a rate that minimizes strain on the crystal.

(13) Before crystal growth, the GaAs polycrystal may be prefabricated by poly synthesis. The poly synthesis process may be performed in a quartz reactor tube utilizing a boat method, for example. Once the polycrystalline GaAs is generated, VGF crystal growth may proceed. To achieve low etch pit density (EPD), several VGF parameters may be carefully controlled. The first parameter may include the shape of the melt/crystal interface **109**, which may be controlled to be concave to the melt, for example, being 5-20 mm concave with the center being 5-20 mm lower than the edge of

the 8 inch diameter crystal. This may be controlled with different temporal and/or spatial temperature profiles.

(14) Second, the crystallization velocity as controlled by the cooling rate may be configured to a range from 0.1-1.0 degrees C./hour. Different solidification rates at different parts of the ingot may be configured. The temperature gradient along the growing crystal can create stress in the boule, where the head cools earlier and in some instances could have the head ~100 degrees cooler than the tail, so precise control is desirable.

(15) Finally, the temperature gradient at the melt/crystal interface may be configured to be between 1-10 degree C./cm. Again, multi-zone heating may enable precise control of the temperature throughout the crucible **130**. After completing the overall solidification, the temperature of the heating may be lowered to room temperature at the speed of about 0.5-5 degrees C./hour, 1-20 degrees C./hour, and 5-20 degrees C./hour, in different heating zones, respectively. This process results in a low-dislocation 8 inch diameter GaAs single crystal of at least 90 mm in length, with a cylindrical body portion. The wafers described here are doped n-type, although p-type and semi-insulating wafers are also enabled by this process. While the process is described for 8 inch diameter ingots, the same process may be used for other diameters (e.g., smaller or larger than 8 inches), with different furnace size. The dislocation density resulting from this process is less than 1000 cm^{-2} , less than 500 cm^{-2} , less than 200 cm^{-2} , less than 100 cm^{-2} , and even less than 30 cm^{-2} , as shown in FIGS. 3A-3D.

(16) In the furnace design shown in FIG. 1, a specific material combination may be selected for the pedestal **140**, so that the crystal ampoule **110** can adjust its position in the furnace according to the growth needs, so as to make the crystal in the desired growth position. The pedestal may move vertically, thereby enabling growth of crystals by vertical Bridgman (VB) method, and also may rotate to improve thermal uniformity in the VGF system **100**. The pedestal **140** and heating coils **120** enable high quality crystal growth along the longitudinal direction. In one example, the crystal loading weight may be greater than 20 kg, and the effective 8-inch crystal length may be 90 mm or higher. This furnace layout enables a VGF process that accurately configures heat transfer and thermal gradients during growth and cooling to maintain appropriate pressure inside the ampoule **110**.

(17) The VGF process disclosed here enables low defect density crystals due to the low thermal gradients that may be maintained during growth. However, even with low gradients, a certain thermal gradient is needed at the solid-liquid interface **109** to maintain continued crystal growth through the entire growth process, and melt temperatures can be high enough to result in high pressures within the ampoule **110**. These higher pressures may be more detrimental as ingot diameters increase, causing deformation of the ampoule **110** itself, which in turn may result in possible deviations from stoichiometry as well as damage to the crystal, crucible, and/or ampoule. To avoid these issues, a multi-zone heating configuration may be utilized, with heat input from the sides and top of the ampoule **110**/crucible **130**, with example heating zones shown by the dashed lines in FIG. 1. In this manner, suitable thermal gradients may be maintained at the growth interface **109** for high quality single crystal growth while still maintaining melt temperatures low enough to reduce and/or eliminate ampoule deformation and resulting issues. In addition, the pedestal **140** enables effective axial heat transfer thereby facilitating single-crystal growth for larger diameters.

(18) FIG. 2 illustrates an 8 inch GaAs ingot and head and tail end faces of the ingot, in accordance with an example embodiment of the disclosure. Referring to the top image, an 8 inch GaAs ingot, or boule, is shown with a length of more than 90 mm, while the lower images show unpolished head-end and tail-end faces of the ingot, showing the diameter uniformity and lack of macroscopic defects such as twins or grain boundaries for the length of the ingot. In instances where there is vertical movement of the pedestal **140**, the furnace design may be “VB-like”, enabling longer crystals with low EPD, where even with movement and VB-like configuration, the crystals may

have lower EPD by appropriate design of temperature profiles, gradients and ramp rates during growth.

(19) The pedestal **140** may move up and down in the vertical direction. First, when the ampoule is set, the position of the ampoule in the heating zone may be adjusted at any time, so that the crystal is in the best temperature field position. Accordingly, this combination method of VGF+VB enables the growth of longer crystals. Compared with the static crucible **130**/ampoule **110**, this approach does not necessarily lead to a higher EPD. The VB+VGF provides an excellent method for growing high-quality, low-dislocation, large-size GaAs crystals.

(20) FIGS. **3A-3D** illustrate etch pit density results for 8 inch gallium arsenide wafers manufactured in accordance with an embodiment of the disclosure. The EPD measurements may be performed in accordance with SEMI M36-0699 and ASTM Test Method F140-92. FIG. **3A** shows EPD results for head-end and tail-end wafers from a first 8 inch ingot, with EPD levels as measured at 69 points (each point having an area of cm^2), with results for the seed (head) and tail ends of the crystal. For this ingot, the wafers show an average EPD of $1288 \text{ cm}^2 \cdot \text{sup.}^{-2}$ at the head end and $71 \text{ cm}^2 \cdot \text{sup.}^{-2}$ at the tail end of the ingot. This is significantly improved over existing GaAs substrates.

(21) FIG. **3B** shows EPD results for head-end and tail-end wafers from a second ingot, with an EPD of $289 \text{ cm}^2 \cdot \text{sup.}^{-2}$ at the head end and $100 \text{ cm}^2 \cdot \text{sup.}^{-2}$ at the tail end, showing the very high quality of the 8 inch GaAs substrates. Hall measurements for the wafers in FIG. **3B** resulted in carrier concentrations ranging from $3 \times 10^{17} \text{ cm}^2 \cdot \text{sup.}^{-3}$ at the head end to $4 \times 10^{18} \text{ cm}^2 \cdot \text{sup.}^{-3}$ at the tail end with resistivity ranging from $5 \times 10^{-3} \Omega \cdot \text{cm}$ at the head end to $2 \times 10^{-3} \Omega \cdot \text{cm}$ at the tail end. Finally, Hall mobility ranged from $3000 \text{ cm}^2/\text{V} \cdot \text{sec}$ at the head end to $2000 \text{ cm}^2/\text{V} \cdot \text{sec}$ at the tail end.

(22) FIG. **3C** shows EPD results for a tail-end wafer (No. 47 of a third boule), where the average EPD is $28 \text{ cm}^2 \cdot \text{sup.}^{-2}$ and max EPD is $250 \text{ cm}^2 \cdot \text{sup.}^{-2}$. Non-contact electrical measurements of this boule showed a carrier concentration ranging from $6 \times 10^{17} \text{ cm}^2 \cdot \text{sup.}^{-3}$ at the head end to $1.8 \times 10^{18} \text{ cm}^2 \cdot \text{sup.}^{-3}$ at the tail end with resistivity ranging from $3.4 \times 10^{-3} \Omega \cdot \text{cm}$ at the head end to $1.7 \times 10^{-3} \Omega \cdot \text{cm}$ at the tail end. Finally, Hall mobility ranged from $2600 \text{ cm}^2/\text{V} \cdot \text{sec}$ at the head end to $2000 \text{ cm}^2/\text{V} \cdot \text{sec}$ at the tail end.

(23) FIG. **3D** illustrates an EPD map and image from an nSpec measurement tool of the wafer of FIG. **3C** (No. 47 of the third boule), where the EPD was $112 \text{ cm}^2 \cdot \text{sup.}^{-2}$ with percentage of wafer with zero EPD was $\sim 50\%$ and the area percentage with $< 1000 \text{ cm}^2 \cdot \text{sup.}^{-2}$ is 97% . The image on the right in FIG. **3D** shows an optical image of the wafer. The nSpec tool measures the entire surface of the wafer as opposed to the EPD map of FIG. **3C**, which is a 69 point measurement.

(24) FIGS. **4A-4F** illustrate photoluminescence measurements of 8 inch gallium arsenide substrates, in accordance with an example embodiment of the disclosure. Referring to FIGS. **4A-4F**, PL intensity maps for different wafers along an 8 inch diameter GaAs boule are shown, showing results for wafers **1, 30, 35, 40, 45, and 47** of the boule.

(25) In the first ~ 10 wafers of the boule, higher intensity signals are evident in four regions situated along 3:00, 6:00, 9:00, and 12:00 o'clock positions around the wafer. This increased intensity may be related to dislocations in these regions at the head-end of the boule. The intensity variation standard deviation is $\sim 20\%$ at the head end, while beyond wafers $\sim 15-20$, the intensity variation is reduced to $\sim 4\%$.

(26) FIGS. **5A-F** illustrate resistivity maps of 8 inch gallium arsenide substrates, in accordance with an example embodiment of the disclosure. High and low values are included in the maps with resistivity values in $\text{m}\Omega \cdot \text{cm}$ and resistivity variation across each wafer standard deviations are shown below each map.

(27) FIGS. **5A-5C** show resistivity plots for three 8 inch GaAs wafers from a boule, where the plots show higher resistivity in the 3:00, 6:00, 9:00, and 12:00 positions around the wafer for the head-end wafers, similar to the PL results of FIGS. **4A-4J**. As with PL intensity, the resistivity variation decreases towards the tail end of the boule, with 12% variation, or $3.4-6.5 \text{ m}\Omega \cdot \text{cm}$, at wafer **5**, and

3%, or 2.1-2.5 mΩ-cm, at wafer **70**.

(28) FIGS. **5D-5F** show resistivity for three 8 inch wafers from the another boule, where the resistivity variation again decreases for wafers closer to the tail-end of the boule. High and low values are shown in the maps ranging from 2.7 to 3.7 mΩ-cm at the head end and 1.7-1.9 mΩ-cm at the tail end, corresponding to a variation standard deviation shown below each, ranging from 9% for wafer **1** to 3.6% for wafer **47**.

(29) FIG. **6** illustrates a method for fabricating GaAs wafers using the vertical gradient freeze furnace **100** shown in FIG. **1**. The process results in 8 inch GaAs substrates with average etch pit densities of less than 1000 cm.^{sup.}-2, less than 500 cm.^{sup.}-2, less than 100 cm.^{sup.}-2, and less than 30 cm.^{sup.}-2 etch pit density. This process may also be used to fabricate indium phosphide (InP), gallium phosphide (GaP) or other related III-V compound semiconductors.

(30) In step **601**, raw arsenic (As), gallium (Ga) and a dopant may be obtained and testing may be performed on the raw materials before being placed in a boat in a quartz tube for poly synthesis to produce doped or undoped polycrystalline GaAs. In some examples, the dopant is silicon. In some examples, the dopant includes one or more types of dopant. In some examples, the dopant is optional. A charge is then prepared by placing the polycrystalline GaAs prefabricated by poly-synthesis in a crucible along with a seed crystal, appropriate amount of B.sub.2O.sub.3 encapsulant, appropriate amount of dopant, evacuating and sealing the crucible inside a fused quartz ampoule.

(31) Once the GaAs charge is generated, vertical gradient freeze (VGF) crystal growth occurs in step **603** where the ampoule with the crucible is heated inside a multi-zone heating system in a controlled manner to progressively melt the polycrystalline charge material containing an optional dopant from the top down until a portion of the seed crystal is melted. Growth may be started from the partially melted seed by implementing a controlled cooling of the multi-zone heater. The temporal and spatial temperature profiles are closely controlled to result low-EPD 8 inch or greater diameter crystals.

(32) To achieve low EPD, several VGF parameters are carefully controlled. The first parameter may include the shape of the melt/crystal interface, which may be controlled to be concave to the melt, for example, being 5-15 mm concave with the center being 5-15 mm lower than the edge of the 8 inch diameter solidified crystal. This may be controlled with different temporal and/or spatial temperature profiles. Second, the crystallization velocity as controlled by the cooling rate may be configured to range from degrees C./hour for different parts of the ingot, while applying a temperature gradient of between 1 and 8 C/cm at the melt-crystal interface. The interface shape may be controlled to be slightly concave to the melt by using appropriate cooling rates in the multi-zone heating system. Different solidification rates at different parts of the ingot may be configured. Finally, the temperature gradient at the melt/crystal interface may be configured to be between 4-8 degree C./cm. Following completion of solidification of the charge material, applying a controlled cooling with appropriate cooling rates of 0.5-5 C/h, 1-10 C/h and 5-20 C/h for different heating zones for the first 300 C, and then at rates of C/h to room temperature may result in very low defect density crystal.

(33) Once the VGF crystals are grown (and optionally tested), in step **605**, an ingot shaping process may be conducted to result in a rounded ingot with desired flats, for example, and may also be tested. Once the ingot is shaped, the ingot may be sliced into wafers in step **607**, and the wafers may be optionally tested.

(34) Once the low EPD wafers have been sliced from the ingot, the wafers may proceed to wafer processing step **609**. An optional wafer annealing process may be performed. In an example annealing process, one or more annealing stages may be used, where the wafers may be loaded vertically into a horizontal quartz boat and inserted in a horizontal quartz ampoule along with arsenic lumps. The arsenic lumps may be configured to provide the needed vapor pressure at the annealing temperature to avoid any arsenic dissociation from the GaAs substrates. The ampoule is

then pumped down to a high vacuum level ($<5 \times 10^{-3}$ Torr) and sealed. The ampoule and its contents may then be inserted into a horizontal 3-zone furnace, for example, and the heating of the ampoule and its contents to the desired set (platform) temperature is initiated.

(35) Testing may be performed on the wafers to ensure structural and electrical quality. In an example scenario, structural quality may be assessed by x-ray characterization and etch pit density measurements, where x-ray transmission measurements may be utilized to assess the presence of slip dislocations. Average dislocation etch pit density (EPD) of less than 1000 cm^{-2} , less than 500 cm^{-2} , less than 200 cm^{-2} , less than 100 cm^{-2} , and even less than 30 cm^{-2} etch pit density, may result in 8 inch GaAs substrates using the above-described process. Structural and electrical quality may also be assessed by photoluminescence and Hall measurements.

(36) Once the low EPD wafers are annealed and optionally tested, a wafer polishing process may be performed that polishes the low EPD wafers and the polished wafers may again be optionally tested. Once the wafers are polished, they may be cleaned and then proceed to step **611** where they may be packaged for shipping to customers. The above processes may also be used to produce indium phosphide (InP) or other compound semiconductor wafers. As a result of this process, low EPD 8 inch GaAs wafers are produced.

(37) FIG. 7 illustrates devices fabricated on low etch pit density 8 inch gallium arsenide wafers, in accordance with an example embodiment of the disclosure. Referring to FIG. 7, there is shown an 8 inch GaAs wafer **701** manufactured using the process described above, and further processed with electrical and/or optoelectronic devices on the die **703**. There is also shown a flat **705**, which may be formed in wafer fabrication to indicate a crystal plane, for example. In another example scenario, a notch may be utilized instead of a flat. The number of die **703** on the wafer **701** may be defined by the area of each die and the size shown in FIG. 7 is merely an example.

(38) In addition, the die **703** may comprise optoelectronic devices such as edge-emitting lasers, vertical cavity surface-emitting lasers (VCSELs), light-emitting diodes (LEDs) including micro-LEDs, and photodetectors where it may be desired to communicate optical signals through the wafer **701** in some scenarios. For example, an array of detectors or surface emitting lasers may be formed on the wafer **701** and readout or control circuitry may be placed on the devices while the optical signals are received and/or transmitted through the wafer **701**.

(39) The process described above may be utilized to fabricate doped substrates, such as silicon-doped substrates, for optoelectronic applications, for example. Silicon, for example, may be incorporated during crystal growth, although other dopants are possible. Dislocations in the substrate can reduce reliability of the devices as current can increase defect size and defects may impact PL, as shown above. In instances where smaller devices are to be fabricated on the substrates, such as micro-LEDs, any defects may comprise a larger area of the device, adversely impacting performance, indicating the advantage of the low EPD crystals disclosed here.

(40) In addition, the process may also be used to generate semi-insulating or low-doped substrates. Semi-insulating GaAs substrates enable high-speed electronic devices due to the high resistivity of the substrates, where conductive substrates can cause losses and parasitic capacitance. Example devices include pseudomorphic high electron mobility transistors (pHEMTs) and heterojunction bipolar transistors.

(41) In an embodiment of the disclosure, wafers for low etch pit density gallium arsenide crystals with silicon dopant (and/or other suitable dopants) may include a gallium arsenide single crystal wafer having an etch pit density of less than 1000 cm^{-2} , less than 500 cm^{-2} , less than 200 cm^{-2} , less than 100 cm^{-2} , and even less than 30 cm^{-2} . The wafer may have a diameter of 8 inches or greater. The wafer may have a dopant concentration in the range of 3×10^{17} to $4 \times 10^{18} \text{ cm}^{-3}$. The wafer may have a thickness of $500 \mu\text{m}$ or greater. Photoluminescence measurements may result in a peak intensity standard deviation of less than 5%.

(42) In another embodiment of the disclosure, a method for low etch pit density gallium arsenide

crystals may include sealing charge material comprising polycrystalline gallium arsenide seed crystal, B.sub.2O.sub.3 encapsulant, and carbon in a crucible; sealing the crucible in a quartz ampoule; performing a vertical gradient freeze crystal growth process by heating the ampoule using a multi-zone heating system to progressively melt the charge material until a portion of the seed crystal is melted; starting growth from the partially melted seed by implementing a controlled cooling of the multi-zone heating system; applying a temperature gradient of between 1 and 8 C/cm at a melt-crystal interface; and controlling a shape of the interface to be concave to the melt utilizing cooling rates in the multi-zone heating system to form a solidified gallium arsenide crystal.

(43) The cooling of the multi-zone heating system may be configured at the rate of 0.1.sup.-2 C/h. The crucible may be evacuated before sealing it into the quartz ampoule. The solidified charge material may be cooled at rates of 0.5-5 C/h, 1-10 C/h, and 5-20 C/h for different heating zones for the first 300 C, and then at rates of 20-50 C/h to room temperature. The interface shape may be controlled to be concave to the melt, with the center being 5-15 mm lower than the edge of the solidified crystal. The solidified crystal may have a diameter of 8 inches or more and may have an etch pit density of less than 1000 cm.sup.-2, less than 500 cm.sup.-2, less than 200 cm.sup.-2, less than 100 cm.sup.-2, and even less than 30 cm.sup.-2. The wafer may have a diameter of 8 inches or greater. The wafer may have a dopant concentration in the range of 3×10^{17} to 4×10^{18} cm.sup.-3. The wafer may have a thickness of 500 μ m or greater. Photoluminescence measurements may result in a peak intensity standard deviation of less than 5%.

(44) In disclosed examples, a vertical gradient freeze system for forming gallium arsenide (GaAs) 8 inch substrates having silicon as a dopant, the system includes a crucible to contain a GaAs liquid melt and seed material during a formation process; one or more heating coils arranged in a plurality of heating zones; and a pedestal to move relative to the crucible, the system operable to control heating of the plurality of heating zones and movement of the pedestal to form a single crystal GaAs substrate.

(45) In some examples, an ampule supports the crucible. In examples, the pedestal is operable to move the crucible relative to the heating coils. In examples, the pedestal is operable to rotate relative to the heating coils. In examples, the pedestal is operable to move vertically relative to the heating coils.

(46) In some examples, the heating coils are operable to activate or the pedestal is operable to move to control a shape of an interface between the GaAs liquid melt and crystals to achieve low etch pit density (EPD). In examples, the shape is concave to the GaAs liquid melt. In examples, the shape of the interface is between 5-20 mm concave, such that the center is 5-20 mm lower than an edge of the substrate.

(47) In some examples, a boron trioxide B.sub.2O.sub.3 layer is placed as an encapsulant on the GaAs liquid melt to reduce loss of crystal growth material. In examples, the heating coils are operable to activate or the pedestal is operable to move to control a crystallization velocity as controlled by the cooling rate may be configured to a range from degrees C./hour.

(48) In some examples, the heating coils are operable to activate or the pedestal is operable to move to control a temperature gradient at the GaAs liquid melt/crystal interface may be configured to be between 1-10 degree C./cm.

(49) In examples, insulation is placed on or within the pedestal to promote radial heat flux inward during growth or heat removal during post-growth process and cooling.

(50) In some examples, one or more electronic or optoelectronic devices are formed on a first surface of the surface. In examples, the electronic or optoelectronic devices are one or more of a light-emitting diodes (LEDs), lasers, heterojunction bipolar transistors (HBTs), and pseudo-morphic high-electron mobility transistors (pHEMTs). In examples, the substrate is diced into a plurality of die, such that optical signals from an optoelectronic device of the electronic or optoelectronic devices on a first side of the substrate are communicated out a second side of the

substrate opposite to the first side.

(51) In some examples, the substrate has an etch pit density of less than 200 cm^{-2} . In examples, the substrate has a dopant concentration of $1 \times 10^{19} \text{ cm}^{-3}$ or greater. In examples, the substrate has a thickness of $300 \text{ }\mu\text{m}$ or greater.

(52) In some disclosed examples, method for forming single crystal gallium arsenide substrates, the method including sealing charge material comprising polycrystalline gallium arsenide (GaAs) seed crystal, B.sub.2O.sub.3 encapsulant, and carbon in a crucible; sealing the crucible in a quartz ampoule; performing a vertical gradient freeze crystal growth process by heating the ampoule using a multi-zone heating system to progressively melt the charge material until a portion of the seed crystal is melted; moving a pedestal relative to the crucible, the system operable to control heating of the multi-zone heating system and movement of the pedestal; and implementing controlled cooling of the multi-zone heating system during growth from the partially melted seed to form a single crystal GaAs substrate 8 inch.

(53) In some examples, the method further includes applying a temperature gradient of between 1 and 8 C/cm at a melt-crystal interface.

(54) In some examples, the method further includes controlling a shape of the interface to be concave to the melt utilizing cooling rates in the multi-zone heating system to form a solidified gallium arsenide crystal. In some examples, the shape of the interface is between 5-20 mm concave, such that the center is 5-20 mm lower than an edge of the substrate.

(55) In some examples, moving the pedestal includes moving the crucible relative to the multi-zone heating system. In some examples, moving the pedestal includes rotates the crucible relative to the multi-zone heating system. In examples, moving the pedestal includes moving the crucible vertically relative to the multi-zone heating system.

(56) In some examples, the method further includes controlling the multi-zone heating system or the pedestal movement to control a crystallization velocity as controlled by the cooling rate may be configured to a range from 0.1-2.0 degrees C./hour.

(57) In some examples, the method further includes forming one or more electronic or optoelectronic devices on a first surface of the substrate. In some examples, the electronic or optoelectronic devices are one or more of a light-emitting diodes (LEDs), lasers, heterojunction bipolar transistors (HBTs), and pseudo-morphic high-electron mobility transistors (pHEMTs).

(58) In some examples, the method further includes evacuating the crucible before sealing it into the quartz ampoule. In examples, the method further includes cooling the solidified charge material at rates of 0.5 to 5 C/h, 1 to 10 C/h and 5 to 20 C/h for different heating zones of the multi-zone heating system for the first 300 C, and then at rates of 20-50 C/h to room temperature.

(59) While the invention has been described with reference to certain embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from its scope. Therefore, it is intended that the present disclosure not be limited to the particular embodiments disclosed, but that the present disclosure will include all embodiments falling within the scope of the appended claims.

Claims

1. A method for forming single crystal gallium arsenide substrates, the method comprising: sealing charge material comprising polycrystalline gallium arsenide (GaAs) liquid melt, B.sub.2O.sub.3 encapsulant, and carbon in a crucible; sealing the crucible in a quartz ampoule; performing a vertical gradient freeze crystal growth process by heating the ampoule using a multi-zone heating system to progressively melt the charge material until a portion of a seed crystal is melted, wherein each zone of the multi-zone heating system is enabled to heat the crucible at different temperatures,

and wherein each zone of the multi-zone heating system includes two or more heating coils; moving a pedestal relative to the crucible through three or more zones of the multi-zone heating system, the system operable to control heating of the multi-zone heating system and movement of the pedestal; implementing controlled cooling of the multi-zone heating system during growth from the partially melted seed to form a single crystal 8 inch GaAs substrate; and controlling a shape of the interface to be concave to the melt utilizing cooling rates in the multi-zone heating system to form a solidified gallium arsenide crystal, wherein the shape of the interface is concave, such that the center is 5-20 mm lower than an edge of the substrate.

2. The method according to claim 1, further comprising applying a temperature gradient of between 1 and 8 C/cm at a melt-crystal interface.

3. The method according to claim 1, wherein moving the pedestal moves the crucible relative to the multi-zone heating system.

4. The method according to claim 1, wherein moving the pedestal rotates the crucible relative to the multi-zone heating system.

5. The method according to claim 1, wherein moving the pedestal moves the crucible vertically relative to the multi-zone heating system.

6. The method according to claim 1, further comprising controlling the multi-zone heating system or the pedestal movement to control a crystallization velocity as controlled by the cooling rate may be configured to a range from 0.1-2.0 degrees C./hour.

7. The method according to claim 1, further comprising forming one or more electronic or optoelectronic devices on a first surface of the substrate.

8. The method according to claim 7, wherein the electronic or optoelectronic devices are one or more of a light-emitting diodes (LEDs), lasers, heterojunction bipolar transistors (HBTs), and pseudo-morphic high-electron mobility transistors (pHEMTs).

9. The method according to claim 1, comprising evacuating the crucible before sealing it into the quartz ampoule.

10. The method according to claim 1, comprising cooling the solidified charge material at rates of 0.5 to 5 C/h, 1 to 10 C/h and 5 to 20 C/h for different heating zones of the multi-zone heating system for the first 300 C, and then at rates of 20-50 C/h to room temperature, wherein different heating zones are cooling the charge at a different temperature.

11. The method according to claim 1, further comprising varying the temperature gradient in different zones of the multi-zone heating system by applying different currents to different heating coils.

12. A method for forming single crystal gallium arsenide substrates, the method comprising: sealing charge material comprising polycrystalline gallium arsenide (GaAs) liquid melt, B.sub.2O.sub.3 encapsulant, and carbon in a crucible; sealing the crucible in a quartz ampoule; performing a vertical gradient freeze crystal growth process by heating the ampoule using a multi-zone heating system to progressively melt the charge material until a portion of a seed crystal is melted; moving a pedestal relative to the crucible, the system operable to control heating of the multi-zone heating system and movement of the pedestal; and implementing controlled cooling of the multi-zone heating system during growth from the partially melted seed; and controlling a shape of the interface to be concave to the melt utilizing cooling rates in the multi-zone heating system to form a solidified gallium arsenide crystal, wherein the shape of the interface is between 5-20 mm concave, such that the center is 5-20 mm lower than an edge of the substrate to form a single crystal 8 inch GaAs substrate.

13. The method according to claim 12, further comprising varying the temperature gradient in different zones of the multi-zone heating system by applying different currents to different heating coils.

14. The method according to claim 12, wherein moving the pedestal moves the crucible relative to the multi-zone heating system.

15. The method according to claim 12, further comprising controlling the multi-zone heating system or the pedestal movement to control a crystallization velocity as controlled by the cooling rate may be configured to a range from 0.1-2.0 degrees C./hour.
 16. The method according to claim 12, further comprising forming one or more electronic or optoelectronic devices on a first surface of the substrate.
 17. The method according to claim 12, comprising evacuating the crucible before sealing it into the quartz ampoule.
 18. The method according to claim 12, comprising cooling the solidified charge material at rates of 0.5 to 5 C/h, 1 to 10 C/h and 5 to 20 C/h for different heating zones of the multi-zone heating system for the first 300 C, and then at rates of 20-50 C/h to room temperature, wherein different heating zones are cooling the charge at a different temperature.
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